

Haocheng Wang
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Portfolio & Projects huch64.github.io

EDUCATION

Nanyang Technological University, SG — M.Computing in Applied AI (MCAAI) 2025.10 – 2027.01
GPA 4.80 / 5.0 | Coursework: Python · RL · Agentic AI · AI Algorithms · AI UX & Data Viz · Data Governance (Expected)
Chongqing Jiaotong University, CN — B.Eng. in Architecture 2011.9 – 2017.6
Graduation with Honors [Top 15%] · Merit Student 2015 [Top 6%]

SELECTED PROJECTS

PRISM — Multi-Agent Research Assistant with Sentence-Level Citations [\[Demo\]](#) 2026.5

- Designed 5-agent workflow (Google ADK + Gemini) with sentence-level indexing (\$.para.sent); built reading/interaction frontend and responsible-AI guardrail layer. Generate-then-locate: answer from full paper, then pin every claim to the exact source sentence.
- Principle-based orchestrator (8 rules, 4 routes) with in-context learning on user feedback. Human-on-the-loop: clickable [id], narrative spectrum, agent trace — UI as interpretability layer, not black-box decoration. Symmetric guardrails, 13/13 probes passed.

My Little Bedroom — RL Agent for Spatial Layout Generation [\[Demo\]](#) 2026.5

- Formulated furniture layout as an MDP (~41K actions/step) and shaped a reward from a few architectural rules of thumb — accessibility, circulation, spatial privacy. Built an interactive demo. MaskablePPO reaches $2.4\times$ a random baseline and transfers to unseen rooms with zero demonstrations.
- Agent settled into consistent, transferable placement patterns. Key finding: reward composition shapes layout quality — multiplicative terms enforce quality floors, additive terms express preferences.

The Unasked — Non-Intrusive Depression Screening via Knowledge Distillation 2025.12

- Built three-stage pipeline: classical ML baselines, NN architecture search, knowledge distillation. Recovered Teacher performance (F1 0.87, AUC 0.92) from 10 non-intrusive behavioral markers — no sensitive data collected.
- Student model: +39% F1, +60% recall. Latent-space analysis showed KD transfers Teacher's decision geometry, quietly acting as a regularizer on weak-signal features.

The Marriage Trap — Data Governance — Bias Diagnostics on UCI Census 2026.02

- Conducted EDA, feature engineering, and intersectional bias diagnostics. Identified accuracy paradox (83% accuracy masks 55% recall failure), uncovered "Marriage Trap" — marital status as proxy inflated rejection $9.74\times$ for single minority women.
- Designed experiment framework and visualization. Systematically compared each stage of feature engineering, quantified recall-fairness tradeoff. Progressive pipeline lifted linear model from AUC 0.46 to 0.906, matching Random Forest baseline.

Additional projects:

CA6002 Unfolding London — energy fingerprint visualization, clustered 5,556 households into 4 energy archetypes for governance framework

CA6000 Rush Hour — traffic forecasting, EDA, gap-aware sliding window + cyclic encoding, SimpleRNN, 35.6% RMSE reduction vs LR.

PROFESSIONAL EXPERIENCE

Yanfei Architects, Shanghai — Project Architect · Class 1 Registered Architect (PRC) 2018.6 – 2024.12
2023 Nanjing Excellent Engineering Survey & Design Award, First Prize in Architecture

- Conducted end-to-end schematic design of [Nanjing Xianlin School](#) (62,000 m²), coordinating structural and MEP teams to resolve spatial conflicts under tight regulatory constraints.
- Designed and delivered a modular furniture system for 40+ rooms at Aranya Youyi Bay MCSC Hotel — broke unique requirements into repeatable components, cutting design-to-fabrication turnaround.
- Data-driven spatial research: mapped reservoir water-level zones via GIS + image mining; quantified residential floor-plan patterns by superimposing hundreds of layouts.

Atelier Li Xinggang, CADG, Beijing — Intern · Parametric Modeling · Stadium Renovation 2016.10 – 2017.1

PUBLICATION

[Soft Architectural Robot](#) — PAM Actuator Composite Apparatus and Architectural Implications 2018.1

- Tsinghua University Parametric Design Workshop. Material research and fabrication. Published in CAADRIA 2018.

SKILLS

AI / ML: Python · PyTorch · TensorFlow / Keras · Gymnasium · scikit-learn · Google ADK
Methods: Reinforcement Learning · Knowledge Distillation · Multi-Agent Orchestration · Data Governance · Responsible AI
Tools: Claude Code · Cursor · Git · Rhino + Grasshopper · Adobe Creative Suite
Languages: Mandarin (Native) · English (Professional Working Proficiency)

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作品与项目详情 huch64.github.io

教育背景

南洋理工大学, 新加坡 — 计算机硕士 (应用人工智能, MCAAI) 2025.10 – 2027.01
GPA 4.80 / 5.0 | 课程: Python · 强化学习 · 智能体 AI · AI 算法 · AI 用户体验与数据可视化 · 数据治理 (预计)

重庆交通大学, 中国 — 建筑学工学学士 2011.9 – 2017.6
优秀毕业生 [前 15%] · 2015 年三好学生 [前 6%]

项目经历

PRISM — 多智能体研究助手与句级引用系统 [\[Demo\]](#) 2026.5

- 设计五智能体协作工作流 (5 句级坐标系), 搭建阅读交互前端与 responsible-AI 护栏层。采用先全文作答、后逐句溯源的策略, 解耦生成与检索, 规避传统 RAG 幻觉引用。
- 构建原则化编排器 (8 条规则、4 条路由), 将用户反馈即时注入为 in-context learning 信号。打造人机协同界面: 可点击 \$ 锚点、叙事光谱、实时跟踪, 以 UI 承载可解释性, 并部署对称护栏, 13/13 安全探测全部通过。

My Little Bedroom — 强化学习驱动的空间布局生成 [\[Demo\]](#) 2026.5

- 将家具布置抽象为 MDP, 提取可达性、私密性、动线等建筑先验驱动 reward shaping, 并设计交互式 demo 供人机对照。智能体达到随机基线 2.4 倍, 零示范即可泛化至未见房型。
- 观察到智能体自发习得稳定、可迁移的布局规律。核心发现: 奖励的组合方式塑造布局质量——乘法项划定质量底线, 加法项传达柔性偏好, 组合决定优先级。

The Unasked — 基于知识蒸馏的非侵入式抑郁筛查 2025.12

- 搭建三阶段递进流程: 建立经典 ML 基线 → 搜索神经网络架构 → 实施知识蒸馏。仅凭 10 个非敏感行为指标即恢复全特征教师性能 (F1 0.87, AUC 0.92), 全程无需采集隐私数据。
- 蒸馏后学生模型 F1 提升 39%、召回率提升 60%。通过隐空间分析揭示: 蒸馏所传递的是教师的决策几何结构而非仅标签, 蒸馏本身起到正则化作用。

The Marriage Trap — 数据治理——UCI 人口普查偏差诊断 2026.02

- 完成 EDA、特征工程与交叉偏差诊断。识别准确率悖论 (83% 准确率掩盖 55% 召回失败), 挖掘出 "婚姻陷阱" 效应——婚姻状态沦为能力的代理变量, 致使单身少数民族裔女性误拒率高达 9.74 倍。
- 设计实验框架与可视化方案, 系统对比特征工程各阶段的数据表现, 量化召回率与群体公平性之间的此消彼长。通过渐进式流水线将线性模型从 AUC 0.46 提升至 0.906, 追平随机森林基线。

其他项目: CA6002 Unfolding London — 电力指纹可视化, 将 5,556 户智能电表数据聚类为 4 种用电行为原型, 为治理提供分析框架; CA6000 Rush Hour — 交通预测, 数据整理, gap-aware 滑动窗口采样 + 周期性编码, SimpleRNN, RMSE 相较 LR 降低 35.6%。

工作经历

雁飞建筑事务所, 上海 — 项目建筑师 · 一级注册建筑师 (中国) 2018.6 – 2024.12

2023 年南京优秀工程勘察设计奖, 建筑类一等奖

- 负责 [南京仙林学校](#) (62,000 m²) 全过程方案设计, 统筹结构与机电团队, 在复杂规范约束下解决空间冲突。
- 为阿那亚友谊湾 MCSC 酒店 40 余间客房设计并落地模块化家具系统——将定制需求拆解为可复用标准化组件, 显著压缩设计到制造周期。
- 开展数据驱动的空间研究: 运用 GIS 与社交媒体图像挖掘绘制水库消落带活动图谱; 叠加数百张住宅平面图提炼共性空间模式与成因。

李兴钢建筑工作室-中国建筑设计研究院, 北京 — 实习 · 参数化建模 · 体育场改造 2016.10 – 2017.1

发表

[Soft Architectural Robot](#) — 空气肌肉驱动的多层复合材料装置设计及其建筑应用 2018.1

- 清华大学暑期参数化建筑工作坊。负责材料研究与制作。发表于 CAADRIA 2018。

技能

AI / ML: Python · PyTorch · TensorFlow / Keras · Gymnasium · scikit-learn · Google ADK
方法: 强化学习 · 知识蒸馏 · 多智能体编排 · 数据治理 · Responsible AI
工具: Claude Code · Cursor · Git · Rhino + Grasshopper · Adobe Creative Suite
语言: 中文 (母语) · 英文 (工作熟练)